

Discrete Mathematics

CS1112

Lecture 7

Induction and Recursion

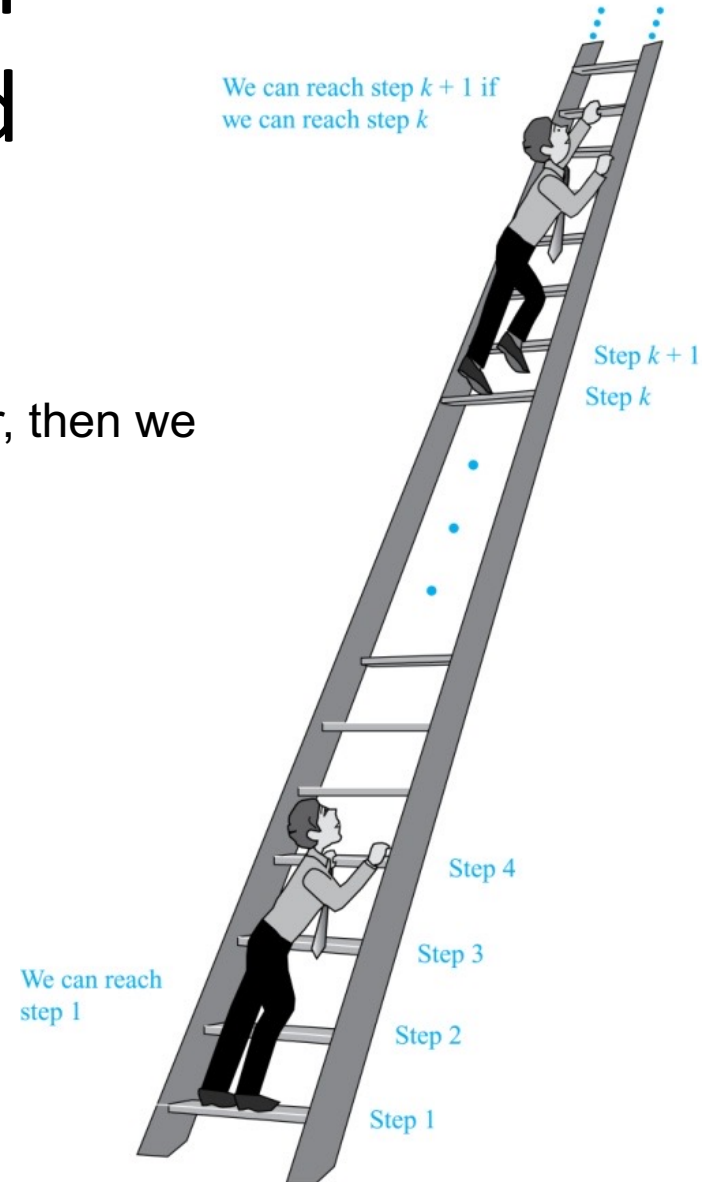
Climbing an Infinite Ladder

Suppose we have an infinite ladder:

1. We can reach the first rung of the ladder.
2. If we can reach a particular rung of the ladder, then we can reach the next rung.

From (1), we can reach the first rung. Then by applying (2), we can reach the second rung. Applying (2) again, the third rung. And so on. We can apply (2) any number of times to reach any particular rung, no matter how high up.

This example motivates proof by mathematical induction.



Principle of Mathematical Induction

Principle of Mathematical Induction: To prove that $P(n)$ is true for all positive integers n , we complete these steps:

- *Basis Step:* Show that $P(1)$ is true.
- *Inductive Step:* Show that $P(k) \rightarrow P(k + 1)$ is true for all positive integers k .

To complete the inductive step, assuming the *inductive hypothesis* that $P(k)$ holds for an arbitrary integer k , show that must $P(k + 1)$ be true.

Climbing an Infinite Ladder Example:

- BASIS STEP: By (1), we can reach rung 1.
- INDUCTIVE STEP: Assume the inductive hypothesis that we can reach rung k . Then by (2), we can reach rung $k + 1$.

Hence, $P(k) \rightarrow P(k + 1)$ is true for all positive integers k . We can reach every rung on the ladder. ◀

Important Points About Using Mathematical Induction

- Mathematical induction can be expressed as the rule of inference

where the domain is the set of positive integers.

- In a proof by mathematical induction, we don't assume that $P(k)$ is true for all positive integers! We show that if we assume that $P(k)$ is true, then $P(k + 1)$ must also be true.
- Proofs by mathematical induction do not always start at the integer 1. In such a case, the basis step begins at a starting point b where b is an integer. We will see examples of this soon.

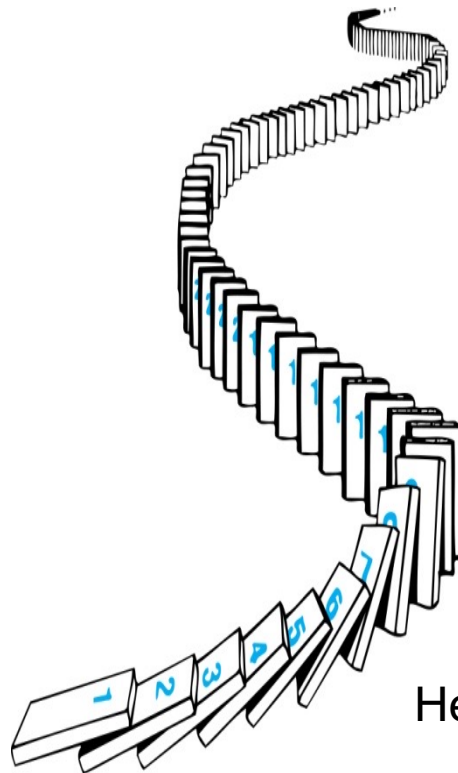
Validity of Mathematical Induction

- Mathematical induction is valid because of the well ordering property, which states that every nonempty subset of the set of positive integers has a least element (*see Section 5.2 and Appendix 1*). Here is the proof:
 - Suppose that $P(1)$ holds and $P(k) \rightarrow P(k + 1)$ is true for all positive integers k .
 - Assume there is at least one positive integer n for which $P(n)$ is false. Then the set S of positive integers for which $P(n)$ is false is nonempty.
 - By the well-ordering property, S has a least element, say m .
 - We know that m can not be 1 since $P(1)$ holds.
 - Since m is positive and greater than 1, $m - 1$ must be a positive integer. Since $m - 1 < m$, it is not in S , so $P(m - 1)$ must be true.
 - But then, since the conditional $P(k) \rightarrow P(k + 1)$ for every positive integer k holds, $P(m)$ must also be true. This contradicts $P(m)$ being false.
 - Hence, $P(n)$ must be true for every positive integer n .

Remembering How Mathematical Induction Works

Consider an infinite sequence of dominoes, labeled $1, 2, 3, \dots$, where each domino is standing.

Let $P(n)$ be the proposition that the n th domino is knocked over.



We know that the first domino is knocked down, i.e., $P(1)$ is true.

We also know that if whenever the k th domino is knocked over, it knocks over the $(k + 1)$ st domino, i.e, $P(k) \rightarrow P(k + 1)$ is true for all positive integers k .

Hence, all dominos are knocked over.

$P(n)$ is true for all positive integers n .

Proving a Summation Formula by Mathematical Induction

Example: Show that:

Solution:

$$\sum_{i=1}^n = \frac{n(n+1)}{2}$$

Note: Once we have this conjecture, mathematical induction can be used to prove it correct.

– BASIS STEP: $P(1)$ is true since $1(1+1)/2 = 1$.

– INDUCTIVE STEP: Assume true for $P(k)$.

The inductive hypothesis is $\sum_{i=1}^k = \frac{k(k+1)}{2}$

Under this assumption,

$$\begin{aligned} 1 + 2 + \dots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2} \end{aligned}$$

Conjecturing and Proving Correct a Summation Formula

Example: Conjecture and prove correct a formula for the sum of the first n positive odd integers. Then prove your conjecture.

Solution: We have: $1 = 1$, $1 + 3 = 4$, $1 + 3 + 5 = 9$, $1 + 3 + 5 + 7 = 16$, $1 + 3 + 5 + 7 + 9 = 25$.

- We can conjecture that the sum of the first n positive odd integers is n^2 ,

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2.$$

- We prove the conjecture is proved correct with mathematical induction.
- BASIS STEP: $P(1)$ is true since $1^2 = 1$.
- INDUCTIVE STEP: $P(k) \rightarrow P(k + 1)$ for every positive integer k .

Assume the inductive hypothesis holds and then show that $P(k + 1)$ holds as well.

Inductive Hypothesis: $1 + 3 + 5 + \cdots + (2k - 1) = k^2$

- So, assuming $P(k)$, it follows that:

$$\begin{aligned} 1 + 3 + 5 + \cdots + (2k - 1) + (2k + 1) &= [1 + 3 + 5 + \cdots + (2k - 1)] + (2k + 1) \\ &= k^2 + (2k + 1) \text{ (by the inductive hypothesis)} \\ &= k^2 + 2k + 1 \\ &= (k + 1)^2 \end{aligned}$$

- Hence, we have shown that $P(k + 1)$ follows from $P(k)$. Therefore the sum of the first n positive odd integers is n^2 .



Proving Divisibility Results

Example: Use mathematical induction to prove that $n^3 - n$ is divisible by 3, for every positive integer n .

Solution: Let $P(n)$ be the proposition that $n^3 - n$ is divisible by 3.

- BASIS STEP: $P(1)$ is true since $1^3 - 1 = 0$, which is divisible by 3.
- INDUCTIVE STEP: Assume $P(k)$ holds, i.e., $k^3 - k$ is divisible by 3, for an arbitrary positive integer k . To show that $P(k + 1)$ follows:

$$\begin{aligned}(k + 1)^3 - (k + 1) &= (k^3 + 3k^2 + 3k + 1) - (k + 1) \\ &= (k^3 - k) + 3(k^2 + k)\end{aligned}$$

By the inductive hypothesis, the first term $(k^3 - k)$ is divisible by 3 and the second term is divisible by 3 since it is an integer multiplied by 3. So by part (i) of Theorem 1 in Section 4.1, $(k + 1)^3 - (k + 1)$ is divisible by 3. 

Therefore, $n^3 - n$ is divisible by 3, for every integer positive integer n .

Recursively Defined Functions

Definition: *A recursive or inductive definition of a function consists of two steps.*

- BASIS STEP: Specify the value of the function at zero.
- RECURSIVE STEP: Give a rule for finding its value at an integer from its values at smaller integers.
- A function $f(n)$ is the same as a sequence $a_0, a_1, \dots$, where a_i , where $f(i) = a_i$. This was done using recurrence relations in Section 2.4.

Recursively Defined Functions

Example: Suppose f is defined by:

$$f(0) = 3,$$

$$f(n + 1) = 2f(n) + 3$$

Find $f(1), f(2), f(3), f(4)$

Solution:

- $f(1) = 2f(0) + 3 = 2 \cdot 3 + 3 = 9$
- $f(2) = 2f(1) + 3 = 2 \cdot 9 + 3 = 21$
- $f(3) = 2f(2) + 3 = 2 \cdot 21 + 3 = 45$
- $f(4) = 2f(3) + 3 = 2 \cdot 45 + 3 = 93$

Example: Give a recursive definition of the factorial function $n!$:

Solution:

$$f(0) = 1$$

$$f(n + 1) = (n + 1) \cdot f(n)$$

Recursively Defined Functions

Example: Give a recursive definition of:

$$\sum_{k=0}^n a_k.$$

Solution: The first part of the definition is

$$\sum_{k=0}^0 a_k = a_0.$$

The second part is

$$\sum_{k=0}^{n+1} a_k = \left(\sum_{k=0}^n a_k \right) + a_{n+1}.$$

Fibonacci Numbers

Example : The Fibonacci numbers are defined as follows:

$$f_0 = 0$$

$$f_1 = 1$$

$$f_n = f_{n-1} + f_{n-2}$$

Find f_2, f_3, f_4, f_5 .

- $f_2 = f_1 + f_0 = 1 + 0 = 1$
- $f_3 = f_2 + f_1 = 1 + 1 = 2$
- $f_4 = f_3 + f_2 = 2 + 1 = 3$
- $f_5 = f_4 + f_3 = 3 + 2 = 5$

In Chapter 8, we will use the Fibonacci numbers to model population growth of rabbits. This was an application described by Fibonacci himself.

Next, we use strong induction to prove a result about the Fibonacci numbers.

Fibonacci Numbers

Example 4: Show that whenever $n \geq 3$, $f_n > \alpha^{n-2}$, where $\alpha = (1 + \sqrt{5})/2$.

Solution: Let $P(n)$ be the statement $f_n > \alpha^{n-2}$. Use strong induction to show that $P(n)$ is true whenever $n \geq 3$.

- BASIS STEP: $P(3)$ holds since $\alpha < 2 = f_3$
 $P(4)$ holds since $\alpha^2 = (3 + \sqrt{5})/2 < 3 = f_4$.
- INDUCTIVE STEP: Assume that $P(j)$ holds, i.e., $f_j > \alpha^{j-2}$ for all integers j with

$3 \leq j \leq k$, where $k \geq 4$. Show that $P(k+1)$ holds, i.e., $f_{k+1} > \alpha^{k-1}$.

- Since $\alpha^2 = \alpha + 1$ (because α is a solution of $x^2 - x - 1 = 0$),
 $\alpha^{k-1} = \alpha^2 \cdot \alpha^{k-3} = (\alpha + 1) \cdot \alpha^{k-3} = \alpha \cdot \alpha^{k-3} + 1 \cdot \alpha^{k-3} = \alpha^{k-2} + \alpha^{k-3}$
- By the inductive hypothesis, because $k \geq 4$ we have

$$f_{k-1} > \alpha^{k-3}, \quad f_k > \alpha^{k-2}.$$

- Therefore, it follows that

$$f_{k+1} = f_k + f_{k-1} > \alpha^{k-2} + \alpha^{k-3} = \alpha^{k-1}.$$

- Hence, $P(k+1)$ is true.

Why does
this equality
hold?



Proofs

Basic proof methods:

- Direct, Indirect, Contradiction, By Cases, Equivalences

Proof of quantified statements:

- **There exists x with some property $P(x)$.**
 - It is sufficient to find one element for which the property holds.
- **For all x some property $P(x)$ holds.**
 - Proofs of ‘For all x some property $P(x)$ holds’ must cover all x and can be harder.
- **Mathematical induction** is a technique that can be applied to prove the universal statements for sets of positive integers or their associated sequences.

Mathematical induction

- Used to prove statements of the form $\forall x P(x)$ where $x \in \mathbb{Z}^+$

Mathematical induction proofs consists of two steps:

- 1) **Basis:** The proposition $P(1)$ is true.
 - 2) **Inductive Step:** The implication $P(n) \rightarrow P(n+1)$, is true for all positive n .
- Therefore we conclude $\forall x P(x)$.
 - **Based on the well-ordering property:** Every nonempty set of nonnegative integers has a **least element**.

Mathematical induction

Example: Prove the sum of first n odd integers is n^2 .

i.e. $1 + 3 + 5 + 7 + \dots + (2n - 1) = n^2$ for all positive integers.

Proof:

- What is $P(n)$? $P(n)$: $1 + 3 + 5 + 7 + \dots + (2n - 1) = n^2$

Basis Step Show $P(1)$ is true

- Trivial: $1 = 1^2$

Inductive Step Show if $P(n)$ is true then $P(n+1)$ is true for all n .

- Suppose $P(n)$ is true, that is $1 + 3 + 5 + 7 + \dots + (2n - 1) = n^2$
- Show $P(n+1)$: $1 + 3 + 5 + 7 + \dots + (2n - 1) + (2n + 1) = (n+1)^2$ follows:

- $$\underbrace{1 + 3 + 5 + 7 + \dots + (2n - 1)}_{n^2} + (2n + 1) = (n+1)^2$$

Mathematical induction

Example: Prove $n < 2^n$ for all positive integers n .

- $P(n)$: $n < 2^n$

Basis Step: $1 < 2^1$ (obvious)

Inductive Step: If $P(n)$ is true then $P(n+1)$ is true for each n .

- Suppose $P(n)$: $n < 2^n$ is true
- Show $P(n+1)$: $n+1 < 2^{n+1}$ is true.

$$\begin{aligned}n + 1 &< 2^n + 1 \\&< 2^n + 2^n \\&= 2^n (1 + 1) \\&= 2^n (2) \\&= 2^{n+1}\end{aligned}$$